



UNIVERSITY OF GENEVA

Faculty of Science

My Course

Digital Forensics

14x065

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1 Introduction

2 Methodology

3 Implementation

4 Results

```
=====
      IMAGE RANGE
=====
dtype: float64
Min: 0.0
Max: 1.0
Range: 1.0
```

Gaussian blur

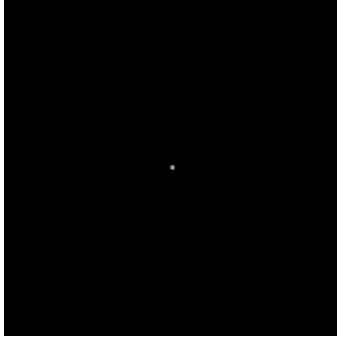
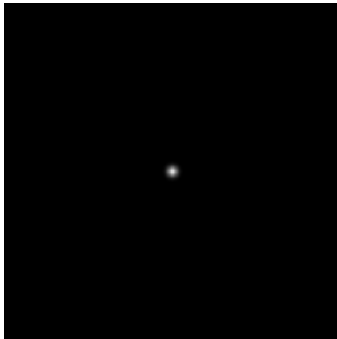
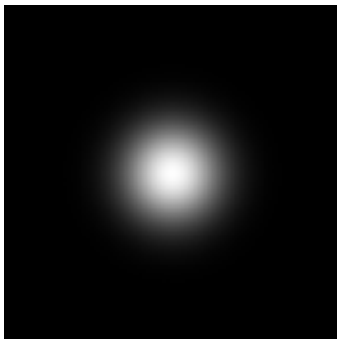
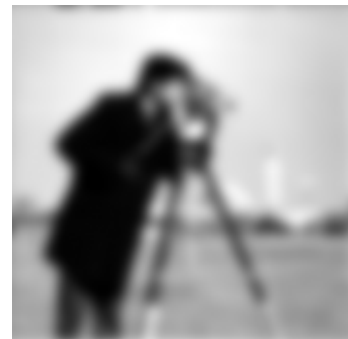
Gaussian kernel, $\sigma = 0.1$ $\sigma = 0.1$, frequency domain $\sigma = 0.1$, Fourier domainGaussian kernel, $\sigma = 1$  $\sigma = 1$, frequency domain $\sigma = 1$, Fourier domainGaussian kernel, $\sigma = 10$  $\sigma = 10$, frequency domain $\sigma = 10$, Fourier domain

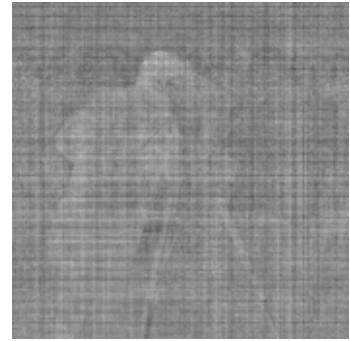
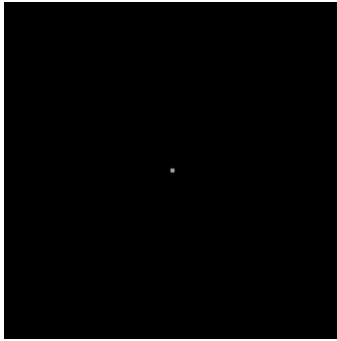
Figure 1: Low Pass Filter

```
=====
      IMAGE RANGE
=====
dtype: float64
Min: 0.0
Max: 1.0
```



```
Range: 1.0
=====
      IMAGE RANGE
=====
dtype: float64
Min: 6.0509799151809614e-24
Max: 1.0
Range: 1.0
=====
      IMAGE RANGE
=====
dtype: float64
Min: 0.024331370730112614
Max: 0.9857897534780872
Range: 0.9614583827479746
=====
      IMAGE RANGE
=====
dtype: float64
Min: 0.03690411600565903
Max: 0.7265833263335318
Range: 0.6896792103278728
```

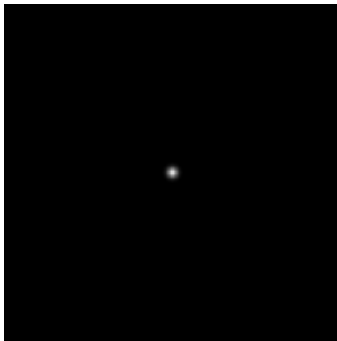
Gaussian kernel, $\sigma = 0.1$ $\sigma = 0.1$, frequency domain $\sigma = 0.1$, Fourier domain



Gaussian kernel, $\sigma = 1$

$\sigma = 1$, frequency domain

$\sigma = 1$, Fourier domain



Gaussian kernel, $\sigma = 10$

$\sigma = 10$, frequency domain

$\sigma = 10$, Fourier domain

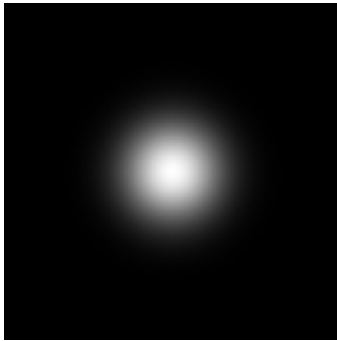


Figure 2: High Pass Filter

```
=====
IMAGE RANGE
=====
dtype: float64
Min: 0.0
Max: 1.0
Range: 1.0
```

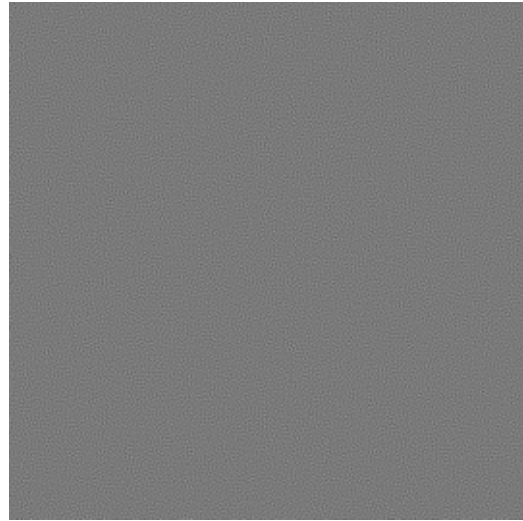
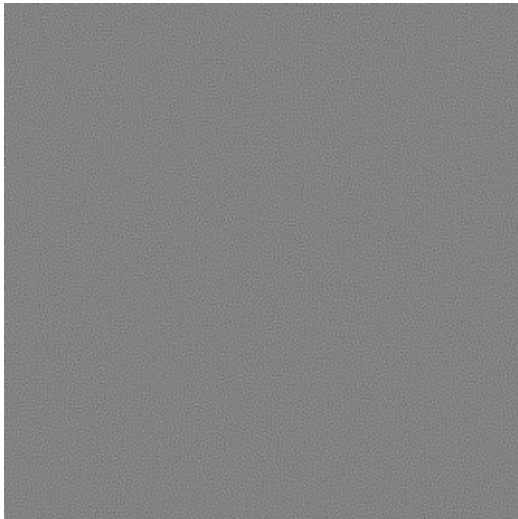
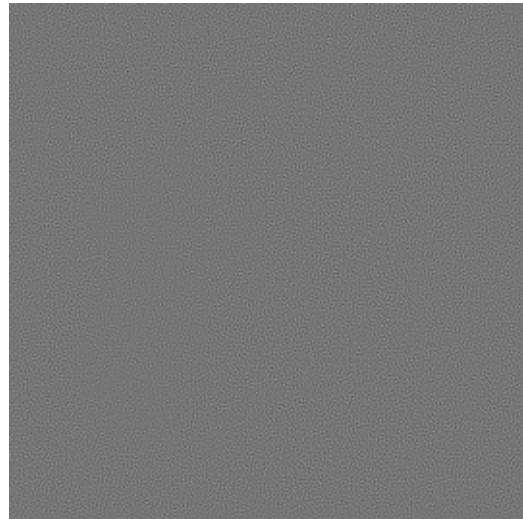
Inverse filtering with noise $\sigma = 0$ Inverse filtering with noise $\sigma = 0.001$ Inverse filtering with noise $\sigma = 0.01$ Inverse filtering with noise $\sigma = 0.1$ 

Figure 3: Inverse Filter

```
=====
IMAGE RANGE
=====
dtype: float64
Min: 0.0
Max: 1.0
Range: 1.0
```

Wiener filtering with noise $\sigma = 0$



Wiener filtering with noise $\sigma = 0.001$



Wiener filtering with noise $\sigma = 0.01$



Wiener filtering with noise $\sigma = 0.1$



Figure 4: Wiener Filter

5 Discussion

Blabla Blabla

6 Conclusion